



## Midterm – MATH 518 – Fall 2025

DATE: Monday, October 20      TIME: 4:05 – 5:25

EXAMINER: Romain Branchereau      TOTAL: 40 points

First name (please write as legibly as possible within the boxes)

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- There are four problems, each worth 10 points, for a total of 40 points.
- Notes, calculators and other electronic devices are not permitted.
- Write your solutions directly on the booklet, under the question (you can use the back of the page).
- You can use any result stated in class or that appeared on the exercise sheets.
- The last two pages can be used as scrap paper and will not be graded.
- Throughout the exam,  $k$  is a field of characteristic 0.



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1. In the following two cases, describe the algebraic set  $V$ , determine the ideal  $I(V)$  and if  $V$  is irreducible .
  - (a) (5 points) The affine algebraic set  $V = V(x^2y^3) \subset k^2$ .
  - (b) (5 points) The projective algebraic set  $V = V(f, g) \subset \mathbb{P}^2(k)$ , where  $f = x^2 + y^2 - z^2$  and  $g = x - z$  in  $k[x, y, z]$ .



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2. (10 points) Find the decomposition of  $V = V(y^2 - xz, z^2 - y^3) \subset k^3$  into irreducibles.



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3. Let  $\varphi: V \longrightarrow W$  be a morphism between two affine varieties  $V \subset k^n$  and  $W \subset k^m$ , and  $\varphi^*: k[W] \longrightarrow k[V]$  the induced  $k$ -algebra morphism.
- (a) (5 points) Show that  $I(\varphi(V)) = \ker(\varphi^*)$ .
  - (b) (5 points) Deduce that  $\varphi^*$  is injective, if and only if  $\varphi(V)$  is dense in  $W$ . *Such a map is called a dominant map.*



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4. (10 points) Show that the affine line  $k$  and the hyperbola  $V = V(xy - 1) \subset k^2$  are not isomorphic as affine varieties.



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